

Reader’s Cheat Sheet for Flock-ZK: A Candidate Zero-Knowledge Extension for Batched BLAKE3

Companion to the implementation-candidate paper

How to use this sheet

This is a plain-language companion to `zk-flock.tex`. It explains the paper’s acronyms, protocol labels, and recurring symbols. It does not add a security claim or replace the paper’s definitions and assumptions.

The shortest useful mental model is:

Part	What it does
BLAKE3 relation	States which hash computations the prover claims to know.
R1CS	Rewrites those computations as algebraic constraints.
Zerocheck	Checks that the multiplication constraints hold.
Lincheck	Connects those constraints to one committed computation trace.
Ring switching	Bridges the bit-level trace and the packed extension-field PCS.
Ligerito PCS	Commits to the trace and proves selected polynomial evaluations.
Merkle and RO	Bind the commitments and derive unpredictable challenges.
ZK masks	Randomize every witness-dependent value the verifier can observe.
Simulator	Produces the same kind of accepted view from the public statement alone.
Extractor	Recovers a witness from a successful prover in the knowledge analysis.

Core cryptography acronyms

ZK: Zero knowledge. A proof is ZK when it reveals nothing about the secret witness beyond the truth of the public statement. In this paper the claim is computational ZK in the classical programmable random-oracle model, for two exact BLAKE3 profiles.

R1CS: Rank-1 Constraint System. An algebraic circuit format whose constraints have the form $(Az) \circ (Bz) = Cz$. The vector z contains the encoded computation trace, and A , B , and C describe the circuit.

IOP: Interactive Oracle Proof. A proof protocol in which the verifier receives large prover messages as oracles and reads only selected locations instead of reading every message in full.

PIOP: Polynomial Interactive Oracle Proof. An IOP whose large messages are represented by polynomials and checked through algebraic identities. Flock’s zerocheck and lincheck form the PIOP layer that checks the R1CS computation.

PCS: Polynomial Commitment Scheme. A mechanism for committing to a polynomial or vector and later proving that selected evaluations are correct without revealing the entire object. Flock uses the Ligerito PCS with ring switching.

FS: Fiat–Shamir. The transformation that replaces an interactive verifier’s random challenges with hashes of the transcript. It makes the proof noninteractive, but its security analysis depends on the random-oracle model.

RO: Random oracle. An idealized random function that returns a consistent, independent random output for each new input. The paper counts the protocol’s oracle calls and includes programming, collision, Merkle-sibling, and grinding-tail terms in its final bound.

ROM: Random Oracle Model. The security model in which the hash function is analyzed as a random oracle. The simulator may program an output only at a fresh input that the adversary has not already queried.

QROM: Quantum Random Oracle Model. The version of the ROM in which a quantum adversary may query a superposition of many inputs at once. QROM does not automatically turn a classical ROM proof into a post-quantum proof. The protocol needs a separate QROM analysis, and its other assumptions and concrete primitives must also withstand quantum attacks. This paper provides no such analysis and makes no post-quantum theorem.

DRBG: Deterministic Random Bit Generator. A generator that expands one unpredictable seed into many pseudorandom values. The prover derives independently labelled mask streams and the proof nonce from a per-proof DRBG.

PoW: Proof of work. The grinding step that searches for a nonce satisfying a hash condition. In the source and transcript framing it appears as `ROLE_POW`.

Algorithms, formats, and implementation terms

BLAKE3: Name of the hash function used by the certified relations. It is not expanded as an acronym. The paper proves batches of its compression function and a fixed-digest relation for exactly 64-byte messages.

SHA-256: Secure Hash Algorithm with a 256-bit output. The implementation uses injectively framed SHA-256 calls for Fiat–Shamir, Merkle nodes, and proof of work. The paper models these calls as one random oracle; it does not prove a standard-model theorem about SHA-256 itself.

IV: Initialization Vector. The fixed starting state of a hash computation. The fixed-digest BLAKE3 circuit pins the IV and the other compression parameters so that the relation means the intended 64-byte BLAKE3 hash.

NTT: Number-Theoretic Transform. A fast linear transform used to encode polynomial data efficiently. This implementation uses an additive NTT over a binary extension field.

SIMD: Single Instruction, Multiple Data. A processor technique that performs the same operation on several values at once. The paper mentions four-way SIMD hashing checked against a scalar reference.

L0: Level zero. The initial PCS codeword layer, before recursive folding. It is pronounced “level zero,” not “ell-zero norm.” The L0 query count and mask dimension control the remaining entropy of unopened leaves.

The proof algorithms at a high level

Zerocheck: Checks the nonlinear R1CS constraints. From the assignment z , define $a = Az$, $b = Bz$, and $c = Cz$. A valid assignment satisfies $a(i)b(i) = c(i)$ at every constraint position i . Zerocheck proceeds in three conceptual steps:

1. interpret the residuals $a(i)b(i) - c(i)$ as evaluations of a polynomial on the Boolean cube;
2. use verifier challenges to fold the claim one variable at a time; and
3. check the final folded claim using evaluations of a , b , and c at one transcript-derived point.

If any residual is nonzero, a false polynomial identity can survive the random folds only on a small exceptional set of challenges. Flock-ZK masks the round messages so those messages do not reveal the witness.

Lincheck: Checks that the linear images came from the committed trace. Zerocheck would be insufficient if a prover could invent unrelated vectors a , b , and c that happen to satisfy $a \circ b = c$. Lincheck prevents this:

1. the verifier randomly combines the public matrix relations $a = Az$, $b = Bz$, and $c = Cz$;
2. the prover supplies a partial evaluation of the committed assignment z ; and
3. the check is reduced to polynomial openings of that same committed z .

Zerocheck handles multiplication; lincheck establishes that the multiplied values are the correct linear functions of the committed computation.

Ligero: The code-based ancestor used by Ligerito. Ligero reshapes data as a matrix, error-correcting-code encodes it, commits to the encoded rows, and opens a random sample of rows. A false matrix relation must corrupt many encoded rows, so random sampling detects it with high probability. It is transparent: it requires no trusted setup.

Ligerito: A recursive polynomial commitment based on Ligero. Ligerito uses the Ligero-style encoded-matrix check as its basic consistency test, combines it with partial sumcheck, and recursively reduces a large polynomial-evaluation claim to smaller ones. This preserves a fast, parallel code-based prover while making the proof substantially smaller than a single plain Ligero matrix opening. Flock uses a Ligerito-derived PCS with Reed–Solomon codes, additive NTT encoding, and Merkle commitments.

Ring switching: Bridges the fields used by the circuit and the PCS. The BLAKE3 computation is naturally a polynomial over bits in $\mathbb{F}_2$, while Flock packs 128 bits into one element of $\mathbb{F}_{2^{128}}$ for efficient commitment. Ring switching reduces an evaluation claim about the original bit-valued polynomial to linked claims about the packed extension-field polynomial. The linking check prevents the prover from treating a packed word as an unrelated field element. It changes the representation used for opening; it does not change the BLAKE3 relation being proved.

Security-model shorthand

Classical ROM: Classical Random Oracle Model. The attacker makes ordinary classical oracle queries. The simulator may program fresh oracle points. This is the model of the paper’s ZK theorem.

Standalone FS: Standalone Fiat–Shamir. The proof derives all challenges from its own transcript, with no external randomness beacon. This is the standalone implementation profile whose knowledge reduction is capped at about 55.994 bits when $Q_H = 2^{64}$.

Knowledge soundness: The accepting prover knows a witness. This is separate from ZK. ZK protects witness privacy; knowledge soundness says that an accepted proof cannot normally be produced without a valid witness.

Simulation extractability: Extraction after simulated proofs. The implemented claim is the weaker fresh-prefix form: extraction is attempted only when the statement, proof nonce, and protocol prefix were not used by the simulator before.

Symbols used repeatedly

Symbol	Meaning
$\mathbb{F}_2$	The field with two elements, represented by bits 0 and 1.
$\mathbb{F}_{2^{128}}$	A 128-bit binary extension field. Most protocol challenges and algebraic masks live here.
$\mathcal{R}_{\text{fd}}$	The fixed-digest relation: 256 public 32-byte digests y_i and secret 64-byte messages x_i satisfying $y_i = \text{BLAKE3}(x_i)$.
Q_H	The adversary’s maximum number of random-oracle queries. The paper evaluates concrete bounds at $Q_H = 2^{64}$.
Q_P	The protocol’s oracle-query bound for one simulated proof and verification, including the analytic grinding cap. The registered value is 997,858.
Q_T	The total oracle budget $Q_H + Q_P$ used in collision and Merkle-boundary terms.
Adv_{ZK}	The distinguisher’s advantage in telling an honest proof from a simulated proof. Smaller is better.
ε_{ZK}	The complete classical-ROM distinguishing bound, including algebraic, programming, Merkle, collision, and grinding-tail terms.
κ_{Γ}	The interactive knowledge-error term associated with the L0 fold access structure.
$G_0, \dots, G_5$	The sequence of hybrid games that changes an honest proof into a simulated proof one controlled step at a time.
z	The packed R1CS assignment or computation trace. It contains witness and intermediate values.
A, B, C	The three R1CS matrices in $(Az) \circ (Bz) = Cz$.
P	The field-valued zerocheck mask polynomial.
Q^*	A public, protocol-fixed affine polynomial multiplied by P in the zerocheck mask. It is not a secret mask or a commitment.

Symbol	Meaning
S, S_c, S_h	Independent mask sources for the lincheck and the first zerocheck round.
μ, g	PCS hiding randomness: μ masks the low message half and g is the full-support blinder.
c	The post-commitment challenge that combines the message and blinder as $F = (\mu \parallel z) + cg$.
γ	A verifier challenge that scales the zerocheck mask.
ρ	The final zerocheck evaluation point.
m	The logarithm of the main packed-vector size used by the PCS/R1CS shape. The registered shape has $m = 22$.
$d_{\log}$	The logarithm of the translated subcube supporting the zerocheck mask. The registered value is 12.

Protocol and code labels

CHUNK_START, CHUNK_END, ROOT: BLAKE3 flags. They specify that the single 64-byte block is both the start and end of a chunk and that the output is the root hash. They are part of the fixed-digest relation, not optional prover settings.

ROLE_POW: Random-oracle role tag for proof of work. Role tags keep proof-of-work inputs separate from transcript and Merkle inputs.

Units

Short form	Meaning
KiB	Kibibyte, exactly $2^{10} = 1,024$ bytes. Unlike KiB, kB conventionally denotes 1,000 bytes.
GB	Gigabyte. In the hardware description it reports installed memory.
ms	Millisecond, one thousandth of a second.
μs	Microsecond, one millionth of a second.
ns	Nanosecond, one billionth of a second.

The two numbers readers most often confuse

Property	Paper’s figure	What it means
Zero knowledge	Complete registered bound: 118.499 bits at $Q_H = 2^{64}$.	Difficulty of distinguishing an honest proof from a simulated proof in the classical programmable ROM.
Knowledge security	Reduction ceiling: at most 55.994 bits before other terms.	Strength of the available theorem that an accepting prover knows a witness. This is not the ZK privacy bound.

The paper does not establish QROM security, a standard-model theorem, or a 100-bit standalone knowledge theorem. It covers only the two registered BLAKE3 profiles and deliberately rejects every uncertified configuration.